INTRODUCTION TO INFORMATION GEOMETRY

Relative Entropy as a Lyapunov Function

Week 6 · Lecture 2

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A scalar function can monitor a flow

For $x' = F(x)$, a Lyapunov function $V(x)$ has a strict minimum at an equilibrium q. If

$$\dot{V}(x) = V'(x)F(x) \leq 0,$$

its value cannot increase along trajectories. When the derivative is strictly negative away from q, the function can establish attraction.

KL divergence from an equilibrium law

For a two-type law, fix $q \in (0,1)$ and define

$$V(x) = D(q\|x)$$

$$= q \ln \frac{q}{x} + (1-q) \ln \frac{1-q}{1-x}.$$

Its minimum is at $x = q$, where $V(q) = 0$.

The key derivative

Differentiate with respect to the evolving coordinate x:

$$V'(x) = -\frac{q}{x} + \frac{1-q}{1-x}$$
$$= \frac{x-q}{x(1-x)}.$$

The denominator will cancel the replicator factor $x(1-x)$.

A global stability calculation

Consider $x' = x(1-x)(3-4x)$ with $q = \frac{3}{4}$. Let $V(x) = D(q\|x)$. Using

$$V'(x) = \frac{x-q}{x(1-x)},$$

find $\dot{V} = V'(x)x'$.

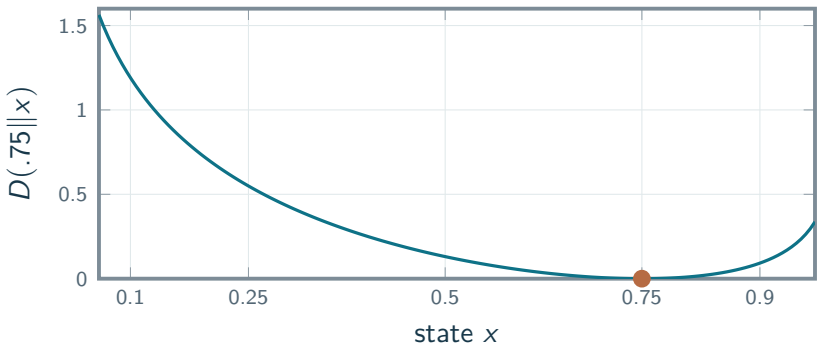
A global stability calculation

Consider $x' = x(1-x)(3-4x)$ with $q = \frac{3}{4}$. The factors cancel:

$$\dot{V} = (x - \tfrac{3}{4})(3 - 4x) = -4(x - \tfrac{3}{4})^2.$$

It is negative at every non-equilibrium interior state.

Relative entropy is a bowl around the target



Its sublevel intervals remain away from 0 and 1.

A global conclusion from local information

For $x' = x(1-x)(a-bx)$ with $b > 0$ and $q = a/b \in (0,1)$,

$$\frac{d}{dt} D(q\|x) = -b(x-q)^2.$$

Because KL diverges at both endpoints, its finite sublevel sets are compact inside $(0,1)$.

Every interior trajectory approaches q.

Many strategies: a short formula

Let $\mathbf{q}$ be a fixed positive reference law and $p_i' = p_i(f_i - \bar{f})$. Differentiate $D(\mathbf{q}\|\mathbf{p}(t))$:

$$\begin{aligned}\frac{d}{dt} D(\mathbf{q}\|\mathbf{p}) &= -\sum_i q_i \frac{p_i'}{p_i} \\ &= \bar{f} - \sum_i q_i f_i .\end{aligned}$$

The cyclic family revisited

For the cyclic family, let $\mathbf{q} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $P = p_A p_B p_C$. We know

$$D(\mathbf{q}\|\mathbf{p}) = -\frac{1}{3}\ln(27P), \qquad \frac{d}{dt}\ln P = (1-s)(1-3Q).$$

Find $\frac{d}{dt}D(\mathbf{q}\|\mathbf{p})$.

The cyclic family revisited

For the cyclic family, let $\mathbf{q} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $P = p_A p_B p_C$. The KL derivative is

$$\frac{d}{dt} D(\mathbf{q} \| \mathbf{p}) = -\frac{1-s}{3}(1 - 3Q).$$

It decays for $s < 1$, is constant for $s = 1$, and increases for $s > 1$.

The quadratic approximation of KL

For a Bernoulli reference q and a small displacement u,

$$D(q\|q+u) = \frac{u^2}{2q(1-q)} + O(u^3).$$

The coefficient depends on the reference distribution. An equal coordinate change can have a different information cost at a different state.

Two equal changes with unequal costs

Use $u = 0.01$ in the quadratic KL approximation

$$D(q\|q+u) \approx \frac{u^2}{2q(1-q)}.$$

Compare $q = 0.2$ with $q = 0.5$.

Two equal changes with unequal costs

Use $u = 0.01$ in the quadratic KL approximation

$$D(q\|q+u) \approx \frac{u^2}{2q(1-q)}.$$

At $q = 0.2$, the cost is 0.0003125. At $q = 0.5$, it is 0.0002.

A quadratic form on the simplex

For an interior reference law $\mathbf{q}$ and tangent displacement $\mathbf{u}$ with $\sum_i u_i = 0$,

$$D(\mathbf{q}\|\mathbf{q}+\mathbf{u}) = \frac{1}{2}\sum_i \frac{u_i^2}{q_i} + O(\|\mathbf{u}\|^3).$$

The quadratic term defines a state-dependent inner product on the tangent space. That is the Fisher information metric.